

Correspondence: Compartment Mapping to Dia Visual Properties

Source feature: Compartment Mapping
Target feature(s): tagName, selector, object of attrs
Automation type: FullAutomation - COMPLEX
Relation: A GMF Compartment Mapping introduces a dedicated visual region inside a node used to structure and present contained content. In JointJS, a possible equivalent is to materialize that region inside the element's SVG/markup by defining a sub-shape via tagName and selector, and then controlling its appearance through an object in attrs.

Main Strategy:

- [A] take the visual representation compartment property (A) of the compartment mapping
- [A] take the name property (B) of A
- [A] take the parent node mapping (C) of the compartment mapping
- [A] take the C's equivalent Dia Element
- [A] add an object to Visual Structure of C with the tagName and selector being 'rect' and B
- [A] add an object to attrs of Visual Attributes of C with the name being B

Strategies of subcorrespondences

Compartment Mapping to Dia Behaviour

- [A] take all the child references (D) of the children property
- [A] take all the node mappings (E) of D
- [A] take all the E's equivalent Dia Elements (F)
- [A] take all the namespaces (G) in F
- [A] if not exist, add a function called 'canEmbed' to Behaviour of C
- [A] add G to the list inside 'canEmbed'
- [A] if not exist, add a function called 'getCompartmentSelector' to Behaviour
- [A] for every namespace (I) in G, add an if statement to 'getCompartmentSelector' in a way that it compares equality between element type and I, and if true, return B
- [A] if not exist, create a JavaScript file called 'util.js' with the default implementation of 'autoEmbedAll' and 'restrictTranslate'
- [A] call 'autoEmbedAll' wherever needed to bind sub-nodes and the super-node together
- [A] bind 'restrictTranslate' and the restrictTranslate property of paper together
- [M] operators must verify the implementation and binding

Structural overview

